

Contact

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Top Skills

Climate Change
Hydrologic Modeling
Hydroinformatics

Honors-Awards

AGU Student Travel Grant
Asian Institute of Technology
Scholarship and Global Water and
Sanitation Center Scholarship
MoChWo Scholarship

Publications

Evaluating Magnitude Agreement
and Occurrence Consistency of
CHIRPS Product with Ground-Based
Observations over Nepal.

Suspended sediment dynamics in an
urban, mountain catchment in Nepal.

Return level analysis of the
Hanumante river using structured
expert judgment.

Leveraging a Citizen Science
Approach for Rainfall Monitoring:
Evaluating Performance and
Reliability.

Hanik Lakhe

Water Engineer & Hydrologist | Researcher | Hydrological modelling
| Remote Sensing and GIS | AI/ML in Hydrology | Citizen Science |
Sustainable Water Resources Management | AIT Graduate Student |
AGU Presenter

Khlong Luang district, Pathum Thani, Thailand

Summary

I'm a Water Engineering and Management graduate student at the Asian Institute of Technology (AIT), Thailand, with 5+ years of research experience spanning hydrology, geospatial analysis, citizen science, and water resource management.

My research focuses on:

- Hydrological modelling (SWAT, HEC-HMS, MIKE+) for water management and flood risk analysis
- # Remote sensing and GIS-based climate risk assessment, water management, CWIS assessments and sanitation infrastructure planning
- # Suspended sediment dynamics in Himalayan river systems using low-cost sensors and citizen science
- AI/ML and big data applications in hydrology and extreme event analysis
- Nature-based solutions (NBS) for urban water management

Currently, I serve as Student Research Assistant at the AIT Global Water & Sanitation Center (GWSC), developing geospatial datasets and ArcGIS/QGIS maps for city-level sanitation assessments across Asia.

As Project Leader of the MESSHI Project (2023–2024), I established a suspended sediment laboratory from scratch, deployed real-time sensors in the Himalayas, managed a citizen science team, and co-authored findings presented at AGU and WaterSciCon24.

Previously, as a Research Associate at Smartphones For Water Nepal (5 years), I contributed to peer-reviewed publications, managed water quality and streamflow datasets, and represented the organization at international conferences.

Publications: 4 peer-reviewed journal papers (EGUsphere, Earth Systems & Environment, Hydrology, Water) + 15+ conference proceedings including AGU Fall Meetings (2019–2023).

MSc Water Engineering & Management — AIT (GPA: 3.87/4.0)
| AGU Student Travel Grant | AIT Scholarship (75%) | GWSC Scholarship (25%) | MoChWo Scholarship

Open to research collaborations, academic positions, and water sector opportunities. | hlakhe123.hl@gmail.com

Experience

AIT Global Water & Sanitation Center

Student Assistant

May 2025 - Present (11 months)

Pathum Thani, Thailand

- Develop and manage geospatial datasets for sanitation infrastructure, urban demographics, and CWIS service coverage across multiple Asian cities using ArcGIS and QGIS.
- Generate thematic maps and spatial analyses to identify infrastructure service gaps and investment priorities for city-level sanitation planning.
- Integrate geo-referenced field surveys and remote sensing data into planning dashboards, improving cross-team data accessibility for development partners.
- Support drafting technical reports and knowledge products communicating GIS outputs to non-technical stakeholders and international organizations.
- Support Center-wide workshops, event preparation, and cross-team documentation aligned with GWSC strategic objectives.

Asian Infrastructure Investment Bank (AIIB)

Climate Risk and GIS Consultant

August 2025 - September 2025 (2 months)

Bangkok City, Thailand

- Conducted a comprehensive climate risk assessment of road infrastructure across 10 provinces in Cambodia, supporting AIIB-funded climate resilience and infrastructure planning initiatives.

- Developed high-resolution multi-hazard risk maps for 38 priority road segments using advanced GIS and spatial analysis techniques.
- Assessed five major climate hazards—flood, drought, landslide, heatwave, and coastal flooding—to evaluate both current and future risk.
- Applied a top-down climate modelling framework using historical baseline data (1995–2014) and future climate projections under SSP2-4.5 and SSP5-8.5 scenarios for the 2040–2059 period.
- Integrated geospatial datasets, climate model outputs, and infrastructure data to analyze identify high-risk roads.
- Designed and implemented a multi-hazard risk classification framework combining multiple factors and projected climate impacts to quantify risk levels.
- Generated clear, decision-ready visualizations and maps to communicate complex climate risks to technical and non-technical stakeholders.
- Delivered geospatial datasets, analytical outputs, and technical documentation to support evidence-based investment planning and climate adaptation strategies.

Water Engineering and Management, AIT
 Graduate Research Assistant | Hydrological Modeller
 January 2025 - June 2025 (6 months)
 Pathum Thani, Thailand

- Developed SWAT hydrological models for water resource assessment and flood risk evaluation in Northern Thailand watersheds.
- Designed a water security framework and identified nature-based solutions (NBS) for urban water management challenges.
- Conducted multi-site case studies on NBS implementation and documented best practices to inform regional water policy.
- Coordinated multi-partner research meetings and served as liaison between field teams and academic supervisors.

Monitoring and Evaluating Suspended Sediment In Himalaya (MESSHI) Project

Project Leader

January 2023 - July 2024 (1 year 7 months)

Lalitpur District, Nepal

- Designed comprehensive project plans, monitoring protocols, and laboratory procedures for suspended sediment analysis across Himalayan river catchments.
- Established a fully operational suspended sediment laboratory from scratch and deployed OBS sensors and data loggers at remote field sites for real-time turbidity data collection.
- Led all fieldwork planning and data collection operations in challenging Himalayan terrain, ensuring rigorous quality control, secure data storage, and meticulous record management.
- Recruited, trained, and managed a team of citizen scientists and volunteers for both field and laboratory activities, expanding monitoring network capacity.
- Coordinated across international stakeholders, research collaborators, and donors to align project milestones and ensure seamless execution throughout the project lifecycle.
- Co-authored findings presented at AGU Fall Meeting 2023 and WaterSciCon24, and published in EGU sphere (2026).
- Prepared comprehensive progress reports and technical documentation, ensuring transparency and accountability to all stakeholders.
- Facilitated workshops and training sessions for local partners, building long-term capacity in suspended sediment monitoring methodologies.

Smartphones For Water Nepal

5 years 1 month

Research Associate

July 2022 - July 2024 (2 years 1 month)

Thasikhel, Lalitpur

- Managed organization-wide water monitoring data with rigorous quality control, validation, and regular dataset updates across citizen science programs.
- Facilitated the Surface Water research group — driving idea development, proposal writing, and implementation of field monitoring initiatives.
- Conducted field data collection, site visits, and performed spatial and statistical data analysis using Python, R, and GIS tools.
- Co-authored peer-reviewed journal papers and conference proceedings
- Recruited, trained, and supervised citizen scientists and volunteers, expanding community-based water monitoring reach across Nepal.
- Represented S4W-Nepal at international and national conferences and stakeholder meetings.
- Managed organizational social media platforms (Facebook, Instagram, YouTube) and led awareness campaigns on water resources.

Research Assistant

January 2020 - June 2022 (2 years 6 months)

Lalitpur District, Nepal

Research Assistant Intern

July 2019 - December 2019 (6 months)

Lalitpur District, Nepal

Himalayan Climate Data Field Lab

Participant

May 2024 - June 2024 (2 months)

Patan, Nepal

- Collaborated with interdisciplinary research groups — including Sensing & Sensors, Chitwan Flooding, AI/Machine Learning, and Re-imagining Resilience and Vulnerability — with professionals from 17+ countries.
- Conducted vulnerability assessments and developed post-disaster management plans for informal settlements along the Bagmati River Corridor using drone and 360° imagery.

- Developed a machine learning model to generate a landslide susceptibility map, enhancing hazard risk analysis.
- Participated in interactive sessions on disaster risk management, resilience building, and climate adaptation.
- Attended workshops on GitHub, Google Earth Engine, and advanced disaster risk modelling techniques.
- Prepared a comprehensive summary report highlighting key learnings and actionable recommendations for field applications.

Young Researchers' Circle

2 years 4 months

Coordinator

August 2021 - July 2022 (1 year)

Thasikhel, Lalitpur

- Led a group of 50+ early-career research enthusiasts, graduate and undergraduate students, fostering collaborative research learning.
- Contributed to the bimonthly Young Researcher newsletter and fortnightly Environmental News Refresher (ENR).
- Organized training programs and seminars to enhance research capacity and engagement among student members.
- Managed financial oversight including budget estimation, fund mobilization, and transparent financial reporting to the executive committee.

Treasurer

April 2020 - August 2021 (1 year 5 months)

Lalitpur District, Nepal

Himalayan Biodiversity and Climate Change Center (HimBioCC)

Research Intern

January 2019 - June 2019 (6 months)

Bhaktapur, Bāgmatī, Nepal

Ecological Stream Health Assessment

Education

Asian Institute of Technology

Master's degree, Water Engineering and Management · (August 2024 - May 2026)

Khwopa College

Bachelor's degree, Environmental Science · (January 2017 - December 2022)

Khwopa Higher Secondary School

+2 Level, Science · (2015 - 2017)

Khwopring English Academy (Secondary)

School Leaving Certificate (SLC) · (2014 - 2015)